Numerical Implementation of Matrix Product States and DMRG
Based on the review article "The density-matrix renormalization group in the age of matrix product states" by U. Schollwöck, Annals of Physics 326, 96–192 (2011)

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I. SCOPE AND CONVENTIONS

These implementation notes summarize the practical coding structure of matrix product state (MPS), matrix product operator (MPO), and finite-system DMRG algorithms for one-dimensional systems with open boundary conditions (OBC). The presentation is based on Schollwöck's review article, but rewritten with explicit implementation conventions suitable for numerical work.

We assume throughout:

- open boundary conditions,
- dense tensors without explicit quantum-number block sparsity,
- finite-system algorithms,
- NumPy-like row-major array storage.

We use the following tensor conventions throughout:

- MPS tensor on site i:

$$A^{[i]}_{\alpha_{i-1},\sigma_i,\alpha_i},$$

  with shape (D_{i-1}, d_i, D_i).

- MPO tensor on site i:

$$W^{[i]}_{\beta_{i-1},\beta_i,\sigma_i,\sigma'_i},$$

  with shape $(\chi_{i-1}, \chi_i, d_i, d_i)$. We use the operator convention

$$\hat{W}^{[i]} = \sum_{\sigma_i,\sigma'_i} W^{[i]}_{\beta_{i-1},\beta_i,\sigma_i,\sigma'_i} |\sigma_i\rangle\langle\sigma'_i|,$$

  so the first physical index is the ket index and the second is the bra index.

- OBC dimensions:

$$D_0 = D_L = 1, \qquad \chi_0 = \chi_L = 1.$$

Unless stated otherwise, all tensors should be stored as `complex128`.

II. RESHAPE CONVENTIONS

For numerical stability and reproducibility, reshape conventions must be fixed once and used everywhere.
For a rank-3 MPS tensor

$$T^{[i]}_{\alpha_{i-1},\sigma_i,\alpha_i},$$

with storage shape (D_{i-1}, d_i, D_i), define the left-grouped matrix

$$T^{[i]}_{(\alpha_{i-1}\sigma_i),\alpha_i} \in \mathbb{C}^{(D_{i-1}d_i)\times D_i}, \tag{1}$$

and the right-grouped matrix

$$T^{[i]}_{\alpha_{i-1},(\sigma_i\alpha_i)} \in \mathbb{C}^{D_{i-1}\times(d_iD_i)}. \tag{2}$$

In row-major (C-order) storage, the flattening used in Eq. (1) is

$$r = \alpha_{i-1}d_i + \sigma_i, \qquad c = \alpha_i. \tag{3}$$

In NumPy:

```
M_left  = T.reshape((D_left * d, D_right), order='C')
M_right = T.reshape((D_left, d * D_right), order='C')
```

Typical inverse reshapes are

```
A  = U.reshape((D_left, d, D_new), order='C')
B  = Vh.reshape((D_new, d, D_right), order='C')
```

All QR/SVD steps must be consistent with these conventions, and the inverse reshape must use the same memory order and grouped-index convention.

III. CANONICAL FORMS

For fixed physical index σ, one may view $A^{[i]\sigma}$ as a matrix from the left bond to the right bond.
Using index notation avoids ambiguity. A tensor $A^{[i]}$ is *left-canonical* if

$$\sum_{\alpha_{i-1},\sigma_i} A^{[i]}_{\alpha_{i-1},\sigma_i,\alpha_i} A^{[i]*}_{\alpha_{i-1},\sigma_i,\alpha'_i} = \delta_{\alpha_i,\alpha'_i}. \tag{4}$$

Equivalently, if $A^{[i]}$ is reshaped as in Eq. (1), then

$$\left(A^{[i]}_{(\alpha_{i-1}\sigma_i),\alpha_i}\right)^\dagger A^{[i]}_{(\alpha_{i-1}\sigma_i),\alpha_i} = \mathbb{I}_{D_i}. \tag{5}$$

A tensor $B^{[i]}$ is *right-canonical* if

$$\sum_{\sigma_i,\alpha_i} B^{[i]}_{\alpha_{i-1},\sigma_i,\alpha_i} B^{[i]*}_{\alpha'_{i-1},\sigma_i,\alpha_i} = \delta_{\alpha_{i-1},\alpha'_{i-1}}. \tag{6}$$

Equivalently, if $B^{[i]}$ is reshaped as in Eq. (2), then

$$B^{[i]}_{\alpha_{i-1},(\sigma_i\alpha_i)} \left(B^{[i]}_{\alpha_{i-1},(\sigma_i\alpha_i)}\right)^\dagger = \mathbb{I}_{D_{i-1}}. \tag{7}$$

A mixed-canonical state with orthogonality center at bond i may be written as

$$|\psi\rangle = \sum_{\alpha_i} s^{[i]}_{\alpha_i} |\alpha_i\rangle_L |\alpha_i\rangle_R, \tag{8}$$

where $s^{[i]}_{\alpha_i} \geq 0$ are Schmidt singular values and the left/right Schmidt states are orthonormal. For a normalized state,

$$\sum_{\alpha_i} \left(s^{[i]}_{\alpha_i}\right)^2 = 1.$$

IV. CANONICALIZATION ALGORITHMS

A. Left sweep by reduced QR

Given a general tensor $M^{[i]}$, reshape it as

$$M^{[i]}_{(\alpha_{i-1}\sigma_i),\alpha_i}.$$

Perform a reduced QR factorization

$$M^{[i]}_{(\alpha_{i-1}\sigma_i),\alpha_i} = QR, \tag{9}$$

where

$$Q \in \mathbb{C}^{(D_{i-1}d_i)\times k}, \qquad R \in \mathbb{C}^{k\times D_i}, \qquad k = \min(D_{i-1}d_i, D_i).$$

Reshape Q into a left-canonical tensor

$$A^{[i]}_{\alpha_{i-1},\sigma_i,\tilde{\alpha}_i},$$

with new bond dimension $\tilde{D}_i = k$, then absorb R into the next tensor:

$$M^{[i+1]}_{\tilde{\alpha}_i,\sigma_{i+1},\alpha_{i+1}} \leftarrow \sum_{\alpha_i} R_{\tilde{\alpha}_i,\alpha_i} M^{[i+1]}_{\alpha_i,\sigma_{i+1},\alpha_{i+1}}. \tag{10}$$

Algorithm 1 LeftCanonicalizeByQR

Require: MPS tensors $\{M^{[1]}, \ldots, M^{[L]}\}$
Ensure: Left-canonical tensors $\{A^{[1]}, \ldots, A^{[L-1]}\}$ and final center tensor $C^{[L]}$

```
for i = 1 to L − 1 do
    reshape M^[i] → M̂ with shape (D_{i−1} d_i) × D_i
    compute reduced QR: M̂ = QR
    reshape Q → A^[i] with shape (D_{i−1}, d_i, D̃_i)
    absorb R into M^[i+1] via Eq. (10)
end for
set C^[L] ← M^[L]
```

a. Important practical note. For canonicalization, use *reduced QR without pivoting.* Pivoted QR changes the bond basis by a permutation and is inconvenient unless the permutation is explicitly propagated into neighboring tensors.

B. Right sweep by reduced QR

To right-canonicalize site i, reshape

$$M^{[i]}_{\alpha_{i-1},(\sigma_i\alpha_i)} \in \mathbb{C}^{D_{i-1}\times(d_iD_i)}.$$

Compute a reduced QR factorization of the Hermitian transpose:

$$\left(M^{[i]}_{\alpha_{i-1},(\sigma_i\alpha_i)}\right)^\dagger = QR, \tag{11}$$

with

$$Q \in \mathbb{C}^{(d_iD_i)\times k}, \qquad R \in \mathbb{C}^{k\times D_{i-1}}, \qquad k = \min(d_iD_i, D_{i-1}).$$

Then

$$M^{[i]} = R^\dagger Q^\dagger. \tag{12}$$

Reshape $Q^\dagger$ into a right-canonical tensor

$$B^{[i]}_{\tilde{\alpha}_{i-1},\sigma_i,\alpha_i},$$

with new bond dimension $\tilde{D}_{i-1} = k$, and absorb $R^\dagger$ into site $i-1$:

$$M^{[i-1]}_{\alpha_{i-2},\sigma_{i-1},\tilde{\alpha}_{i-1}} \leftarrow \sum_{\alpha_{i-1}} M^{[i-1]}_{\alpha_{i-2},\sigma_{i-1},\alpha_{i-1}} (R^\dagger)_{\alpha_{i-1},\tilde{\alpha}_{i-1}}. \tag{13}$$

V. SVD TRUNCATION

Given a matrix $X \in \mathbb{C}^{m\times n}$ with reduced SVD

$$X = USV^\dagger, \tag{14}$$

where

$$U \in \mathbb{C}^{m\times r}, \quad S = \mathrm{diag}(s_1, \ldots, s_r), \quad V^\dagger \in \mathbb{C}^{r\times n}, \quad s_1 \geq s_2 \geq \cdots \geq s_r \geq 0,$$

truncate to bond dimension D by keeping the D largest singular values:

$$X \approx U_D S_D V_D^\dagger. \tag{15}$$

The discarded weight is

$$\eta = \sum_{j>D} s_j^2. \tag{16}$$

If the state is normalized, η is the squared 2-norm of the discarded Schmidt tail.

Algorithm 2 TruncateBySVD

Require: matrix X, max bond dimension $D_{\max}$, tolerance ϵ
Ensure: truncated $U_D, S_D, V_D^\dagger$, discarded weight η

```
compute reduced SVD: X = USV†
let D = min{D_max, #(s_j > ε)}
if D = 0 then
    D ← 1
end if
keep first D singular values/vectors
η ← Σ_{j>D} s_j^2
```

a. Practical note. In production codes, the retained bond dimension is often chosen using both a hard cap $D_{\max}$ and a discarded-weight tolerance, rather than only a singular-value threshold. A robust rule is to retain the smallest $D \leq D_{\max}$ such that the discarded weight is below a prescribed tolerance, while always keeping at least one singular value.

VI. HEISENBERG MPO

For the spin-1/2 nearest-neighbor XXZ chain with field,

$$H = \sum_{i=1}^{L-1} \left[\frac{J}{2} \left(S_i^+ S_{i+1}^- + S_i^- S_{i+1}^+ \right) + J_z S_i^z S_{i+1}^z \right] - h \sum_{i=1}^{L} S_i^z, \tag{17}$$

a standard OBC MPO with bond dimension $\chi = 5$ is

$$\hat{W}_{\text{bulk}} = \begin{pmatrix} \mathbb{I} & 0 & 0 & 0 & 0 \\ S^+ & 0 & 0 & 0 & 0 \\ S^- & 0 & 0 & 0 & 0 \\ S^z & 0 & 0 & 0 & 0 \\ -hS^z & \frac{J}{2}S^- & \frac{J}{2}S^+ & J_z S^z & \mathbb{I} \end{pmatrix}. \tag{18}$$

Here $\hat{W}_{\text{bulk}}$ is an operator-valued matrix. The tensor components $W^{[i]}_{\beta_{i-1},\beta_i,\sigma,\sigma'}$ are obtained by expanding each operator entry in the local basis:

$$(\hat{O})_{\sigma,\sigma'} = \langle\sigma|\hat{O}|\sigma'\rangle.$$

For the local spin-1/2 basis $\{|\uparrow\rangle, |\downarrow\rangle\}$, use

$$S^+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad S^- = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \qquad S^z = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

For OBC, choose:

$$\hat{W}^{[1]} = \begin{pmatrix} -hS^z & \frac{J}{2}S^- & \frac{J}{2}S^+ & J_zS^z & \mathbb{I} \end{pmatrix}, \tag{19}$$

$$\hat{W}^{[L]} = \begin{pmatrix} \mathbb{I} \\ S^+ \\ S^- \\ S^z \\ -hS^z \end{pmatrix}, \tag{20}$$

and use $\hat{W}_{\text{bulk}}$ on all intermediate sites, with the chosen lower-triangular convention.

In code, this means:

- the left boundary tensor has shape $(1, 5, d, d)$ and is the row vector selecting the bottom row of the bulk operator-valued MPO,
- the right boundary tensor has shape $(5, 1, d, d)$ and is the column vector selecting the leftmost column of the bulk operator-valued MPO.

a. Recommended unit test. For $L = 2, 3, 4$, contract the MPO to a dense Hamiltonian and compare entrywise against the Hamiltonian constructed directly by Kronecker products of the local operators.

VII. AKLT HAMILTONIAN MPO

For the spin-1 AKLT chain, the Hamiltonian is

$$H_{\text{AKLT}} = \sum_{i=1}^{L-1} \left[\mathbf{S}_i \cdot \mathbf{S}_{i+1} + \frac{1}{3} (\mathbf{S}_i \cdot \mathbf{S}_{i+1})^2 \right]. \tag{21}$$

This is the Hamiltonian whose exact ground state is the AKLT MPS given later in these notes.

A. Local spin-1 operators

In the local basis

$$\{|+1\rangle, |0\rangle, |-1\rangle\},$$

use the standard spin-1 matrices

$$S^z = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \tag{22}$$

$$S^+ = \sqrt{2}\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \qquad S^- = \sqrt{2}\begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}. \tag{23}$$

Define the bilinear bond operator

$$X_{i,i+1} \equiv \mathbf{S}_i \cdot \mathbf{S}_{i+1} = \frac{1}{2}\left(S_i^+ S_{i+1}^- + S_i^- S_{i+1}^+\right) + S_i^z S_{i+1}^z. \tag{24}$$

Then

$$h_{i,i+1}^{\mathrm{AKLT}} = X_{i,i+1} + \frac{1}{3} X_{i,i+1}^2, \tag{25}$$

and

$$H_{\mathrm{AKLT}} = \sum_{i=1}^{L-1} h_{i,i+1}^{\mathrm{AKLT}}.$$

B. Factorized form of the bond interaction

Introduce the operator lists

$$O_1 = \frac{1}{\sqrt{2}} S^+, \qquad O_2 = \frac{1}{\sqrt{2}} S^-, \qquad O_3 = S^z, \tag{26}$$

and

$$\bar{O}_1 = \frac{1}{\sqrt{2}} S^-, \qquad \bar{O}_2 = \frac{1}{\sqrt{2}} S^+, \qquad \bar{O}_3 = S^z. \tag{27}$$

Then Eq. (24) becomes

$$X_{i,i+1} = \sum_{a=1}^{3} O_a^{[i]} \bar{O}_a^{[i+1]}. \tag{28}$$

Therefore

$$X_{i,i+1}^2 = \sum_{a=1}^{3} \sum_{b=1}^{3} \left(O_a O_b\right)^{[i]} \left(\bar{O}_a \bar{O}_b\right)^{[i+1]}. \tag{29}$$

Substituting Eqs. (28) and (29) into Eq. (25), we obtain

$$h_{i,i+1}^{\mathrm{AKLT}} = \sum_{a=1}^{3} O_a^{[i]} \bar{O}_a^{[i+1]} + \frac{1}{3} \sum_{a=1}^{3} \sum_{b=1}^{3} \left(O_a O_b\right)^{[i]} \left(\bar{O}_a \bar{O}_b\right)^{[i+1]}. \tag{30}$$

C. Exact MPO construction

Equation (30) is an exact nearest-neighbor MPO. A convenient MPO has bond dimension

$$\chi = 1 + 3 + 9 + 1 = 14.$$

This slightly larger bond dimension is chosen because it leads to a completely explicit implementation without block-index ambiguities.

a. Explicit MPO bond-index convention. For implementation, the MPO bond index is ordered as follows:

$$\beta = 0 \qquad \text{start channel}, \tag{31}$$

$$\beta = 1, 2, 3 \qquad \leftrightarrow \qquad a = 1, 2, 3, \tag{32}$$

$$\beta = 4, \ldots, 12 \qquad \leftrightarrow \qquad (a, b) \in \{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3)\}, \tag{33}$$

taken in lexicographic order, and

$$\beta = 13 \qquad \text{terminal / identity-propagation channel.} \tag{34}$$

This convention is used consistently in the bulk tensor and in both boundary tensors.

Define

$$\boldsymbol{O} = \begin{pmatrix} O_1 & O_2 & O_3 \end{pmatrix}, \qquad \bar{\boldsymbol{O}} = \begin{pmatrix} \bar{O}_1 \\ \bar{O}_2 \\ \bar{O}_3 \end{pmatrix}, \tag{35}$$

$$\boldsymbol{O}^{(2)} = \begin{pmatrix} O_1O_1 & O_1O_2 & O_1O_3 & O_2O_1 & O_2O_2 & O_2O_3 & O_3O_1 & O_3O_2 & O_3O_3 \end{pmatrix}, \tag{36}$$

and

$$\bar{\boldsymbol{O}}^{(2)} = \begin{pmatrix} \bar{O}_1\bar{O}_1 \\ \bar{O}_1\bar{O}_2 \\ \bar{O}_1\bar{O}_3 \\ \bar{O}_2\bar{O}_1 \\ \bar{O}_2\bar{O}_2 \\ \bar{O}_2\bar{O}_3 \\ \bar{O}_3\bar{O}_1 \\ \bar{O}_3\bar{O}_2 \\ \bar{O}_3\bar{O}_3 \end{pmatrix}. \tag{37}$$

The ordering of the 9 entries in $\boldsymbol{O}^{(2)}$ and $\bar{\boldsymbol{O}}^{(2)}$ is the same lexicographic ordering defined in Eq. (33).

A bulk operator-valued MPO tensor is then

$$\hat{W}_{\text{AKLT,bulk}} = \begin{pmatrix} \mathbb{I} & \boldsymbol{O} & \frac{1}{3}\boldsymbol{O}^{(2)} & 0 \\ 0 & 0 & 0 & \bar{\boldsymbol{O}} \\ 0 & 0 & 0 & \bar{\boldsymbol{O}}^{(2)} \\ 0 & 0 & 0 & \mathbb{I} \end{pmatrix}, \tag{38}$$

where the block sizes are $1|3|9|1$.

In words:

- the channel $0 \to a$ with $a = 1, 2, 3$ starts the bilinear term via O_a,
- the channel $0 \to (a, b)$ starts the biquadratic term via $\frac{1}{3}O_aO_b$,
- the channel $a \to 13$ finishes the bilinear term via $\bar{O}_a$,
- the channel $(a, b) \to 13$ finishes the biquadratic term via $\bar{O}_a\bar{O}_b$,
- the channel $13 \to 13$ propagates the identity after a term has been completed.

With the index convention above, contracting the MPO generates exactly

$$\sum_{i=1}^{L-1} h_{i,i+1}^{\text{AKLT}} = \sum_{i=1}^{L-1} \left[\mathbf{S}_i \cdot \mathbf{S}_{i+1} + \frac{1}{3} \left(\mathbf{S}_i \cdot \mathbf{S}_{i+1} \right)^2 \right].$$

For OBC, use boundary tensors

$$\hat{W}_{\text{AKLT}}^{[1]} = \begin{pmatrix} \mathbb{I} & \boldsymbol{O} & \frac{1}{3}\boldsymbol{O}^{(2)} & 0 \end{pmatrix}, \tag{39}$$

and

$$\hat{W}_{\text{AKLT}}^{[L]} = \begin{pmatrix} 0 \\ \bar{\boldsymbol{O}} \\ \bar{\boldsymbol{O}}^{(2)} \\ \mathbb{I} \end{pmatrix}. \tag{40}$$

Thus, in the tensor convention of these notes,

- the left boundary tensor has shape $(1, 14, 3, 3)$,
- each bulk tensor has shape $(14, 14, 3, 3)$,
- the right boundary tensor has shape $(14, 1, 3, 3)$.

The MPO physical indices are ordered as

$$W^{[i]}_{\beta_{i-1},\beta_i,\sigma,\sigma'},$$

with σ the ket index and σ' the bra index, exactly as in the Heisenberg MPO section.

D. Tensor components

As in the Heisenberg MPO section, the operator-valued entries are converted into numerical MPO tensor components by

$$W^{[i]}_{\beta_{i-1},\beta_i,\sigma,\sigma'} = \langle\sigma|\hat{W}^{[i]}_{\beta_{i-1},\beta_i}|\sigma'\rangle. \tag{41}$$

Here $\sigma, \sigma' \in \{+1, 0, -1\}$ and the local physical dimension is $d = 3$.

E. Implementation remarks

A transparent implementation strategy is:

1. construct the local spin-1 matrices S^+, S^-, S^z,
2. define O_a and $\bar{O}_a$,
3. build the 9 operators $O_a O_b$,
4. build the 9 operators $\bar{O}_a \bar{O}_b$,
5. assemble the operator-valued boundary and bulk MPOs from Eqs. (38), (39), and (40),
6. expand each operator entry into local matrix elements.

a. Recommended unit tests. For $L = 2, 3, 4$:

- contract the MPO to a dense matrix,
- compare against the Hamiltonian built directly from Eq. (21),
- verify that the AKLT MPS given later in these notes has energy

$$E_0 = -\frac{2}{3}(L-1)$$

  for open chains, up to numerical precision.

VIII. MPO ENVIRONMENTS

For a center at site i, let $L^{[i]}$ denote the contraction of sites $1, \ldots, i-1$ into a left environment, and let $R^{[i]}$ denote the contraction of sites $i+1, \ldots, L$ into a right environment.

We store the left and right environments as tensors

$$L^{[i]}_{\beta_{i-1}}(\alpha_{i-1}, \alpha'_{i-1}), \qquad R^{[i]}_{\beta_i}(\alpha_i, \alpha'_i),$$

with array shapes

$$L^{[i]} \in \mathbb{C}^{\chi_{i-1}\times D_{i-1}\times D_{i-1}}, \qquad R^{[i]} \in \mathbb{C}^{\chi_i\times D_i\times D_i}.$$

A. Left environment update

Suppose sites $1, \ldots, i-1$ are represented by left-canonical tensors. Then the left environment is updated recursively by

$$L^{[i]}_{\beta_{i-1}}(\alpha_{i-1}, \alpha'_{i-1}) = \sum_{\substack{\alpha_{i-2}, \alpha'_{i-2} \\ \sigma, \sigma' \\ \beta_{i-2}}} L^{[i-1]}_{\beta_{i-2}}(\alpha_{i-2}, \alpha'_{i-2}) A^{[i-1]}_{\alpha_{i-2}, \sigma, \alpha_{i-1}} W^{[i-1]}_{\beta_{i-2}, \beta_{i-1}, \sigma, \sigma'} A^{[i-1]*}_{\alpha'_{i-2}, \sigma', \alpha'_{i-1}}. \tag{42}$$

With array conventions

- `Lold[b,x,y]`,
- `A[x,s,a]`,
- `W[b,B,s,t]`,

the validated NumPy implementation is

```
Lnew = np.einsum('bxy,xsa,bBst,ytc->Bac', Lold, A, W, A.conj())
```

B. Right environment update

Suppose sites $i+1, \ldots, L$ are represented by right-canonical tensors. Then the right environment is updated recursively by

$$R^{[i]}_{\beta_i}(\alpha_i, \alpha'_i) = \sum_{\substack{\alpha_{i+1}, \alpha'_{i+1} \\ \sigma, \sigma' \\ \beta_{i+1}}} B^{[i+1]}_{\alpha_i, \sigma, \alpha_{i+1}} W^{[i+1]}_{\beta_i, \beta_{i+1}, \sigma, \sigma'} R^{[i+1]}_{\beta_{i+1}}(\alpha_{i+1}, \alpha'_{i+1}) B^{[i+1]*}_{\alpha'_i, \sigma', \alpha'_{i+1}}. \tag{43}$$

With array conventions

- `Rold[B,a,c]`,
- `B[x,s,a]`,
- `W[b,B,s,t]`,

the validated NumPy implementation is

```
Rnew = np.einsum('xsa,bBst,Bac,ytc->bxy', B, W, Rold, B.conj())
```

C. Boundary initialization

For open boundary conditions,

$$L^{[1]}_{\beta_0}(1,1) = \delta_{\beta_0,1}, \qquad R^{[L]}_{\beta_L}(1,1) = \delta_{\beta_L,1}, \tag{44}$$

with

$$D_0 = D_L = \chi_0 = \chi_L = 1.$$

In code, these are simply

```
L = np.zeros((1,1,1), dtype=np.complex128); L[0,0,0] = 1
R = np.zeros((1,1,1), dtype=np.complex128); R[0,0,0] = 1
```

D. Interpretation and convention warning

The update formulas above define the *stored* left and right environment tensors. However, for the tensor, reshape, and MPO conventions used in these notes, the stored left environment does not enter the local effective-Hamiltonian action with the same bond-index order as it is stored.

More precisely, if the stored left environment is represented in code as

```
L[b,x,y],
```

then in the validated one-site and two-site local contractions it must be used as

```
L[b,y,x].
```

This point is easy to miss. A local operator built from environments can remain Hermitian even when this ordering is wrong, while still failing to agree with the exact projected operator

$$P^\dagger H P.$$

Therefore one should distinguish carefully between

- the recursion defining the stored environments, and
- the way those stored tensors are inserted into the local effective-Hamiltonian action.

E. Practical recommendation

The left and right environment update routines should be tested independently against brute-force contractions on very small systems. After that, the resulting one-site and two-site environment-based local operators should be compared explicitly against dense projected references. In practice, this second test is the decisive one: it verifies not only that the environments are built recursively, but also that they are inserted into the local effective Hamiltonian with the correct index ordering.

IX. ONE-SITE EFFECTIVE HAMILTONIAN

Let the center tensor at site i be

$$M^{[i]}_{\alpha_{i-1},\sigma_i,\alpha_i}.$$

The local variational problem is

$$\mathcal{H}_{\text{eff}}\, v = Ev, \tag{45}$$

where v denotes the vectorized form of the center tensor.

a. General structure. The one-site effective Hamiltonian is obtained by contracting the full MPO with all MPS tensors outside site i, leaving only the local tensor $M^{[i]}$ open. In principle one may write this object symbolically as a contraction of a left environment, one local MPO tensor, and a right environment. In practice, however, the safest way to specify the operator—and the one that is directly useful for coding—is through its validated matrix-free contraction sequence.

b. Convention-sensitive point. For the conventions used in these notes, the stored left environment tensor does not enter the local action with the same MPS-bond index order as it is stored. If the stored array is `L[b,x,y]`, then the validated local contraction uses it as `L[b,y,x]`. Equivalently, the left environment enters the one-site effective Hamiltonian with its two MPS-bond indices interchanged relative to the most naive symbolic reading.

A. Validated matrix-free action

Let

$$L^{[i]}_{\beta_{i-1}}(\alpha_{i-1}, \alpha'_{i-1})$$

be the stored left environment and

$$R^{[i]}_{\beta_i}(\alpha_i, \alpha'_i)$$

the stored right environment. Then the validated one-site matrix-free action is

$$X_{\beta_{i-1},\alpha_{i-1},\sigma_i,\alpha'_i} = \sum_{\alpha'_{i-1}} L^{[i]}_{\beta_{i-1}}(\alpha'_{i-1}, \alpha_{i-1}) M^{[i]}_{\alpha'_{i-1},\sigma_i,\alpha'_i}, \tag{46}$$

$$Y_{\beta_i,\alpha_{i-1},\sigma'_i,\alpha'_i} = \sum_{\beta_{i-1},\sigma_i} W^{[i]}_{\beta_{i-1},\beta_i,\sigma'_i,\sigma_i} X_{\beta_{i-1},\alpha_{i-1},\sigma_i,\alpha'_i}, \tag{47}$$

$$(\mathcal{H}_{\text{eff}} M)_{\alpha_{i-1},\sigma'_i,\alpha_i} = \sum_{\beta_i,\alpha'_i} Y_{\beta_i,\alpha_{i-1},\sigma'_i,\alpha'_i} R^{[i]}_{\beta_i}(\alpha'_i, \alpha_i). \tag{48}$$

This is the convention-correct form for the one-site effective Hamiltonian in the present notes. In particular, the first step involves

$$L^{[i]}_{\beta_{i-1}}(\alpha'_{i-1}, \alpha_{i-1}),$$

not

$$L^{[i]}_{\beta_{i-1}}(\alpha_{i-1}, \alpha'_{i-1}).$$

B. Validated NumPy implementation

With arrays

- `L[b,x,y]` for the stored left environment,
- `M[y,s,z]` for the local center tensor,
- `W[b,B,s,t]` for the MPO tensor,
- `R[B,z,a]` for the stored right environment,

the validated implementation is

```
X  = np.einsum('byx,ysz->bxsz', L, M)
Y  = np.einsum('bBst,bxsz->Bxtz', W, X)
HM = np.einsum('Bxtz,Bza->xta', Y, R)
```

Here `HM` has the same shape as `M`. The use of `'byx'` rather than `'bxy'` in the first contraction is essential.

a. Interpretation of physical indices. The MPO tensor is stored as

$$W^{[i]}_{\beta_{i-1},\beta_i,\sigma,\sigma'},$$

with the first physical index σ the ket/output index and the second physical index σ' the bra/input index. The contraction in Eq. (47) is the validated routing consistent with this convention and with the environment storage described above.

C. Vectorization and local eigensolver interface

If

$$M^{[i]} \in \mathbb{C}^{D_{i-1} \times d_i \times D_i},$$

then the local vector-space dimension is

$$N_{\text{loc}} = D_{i-1} d_i D_i.$$

The eigensolver vector

$$v \in \mathbb{C}^{N_{\mathrm{loc}}}$$

is identified with the tensor by

$$M^{[i]} = \mathrm{reshape}(v,\, (D_{i-1}, d_i, D_i)),$$

using the fixed C-order convention of Section II. After applying the local effective Hamiltonian, the output tensor is flattened back into a vector with the same convention.

A practical wrapper is

```
from scipy.sparse.linalg import LinearOperator, eigsh

def heff_one_site_matvec(v, L, W, R, Dl, d, Dr):
M = v.reshape((Dl, d, Dr), order='C')
X  = np.einsum('byx,ysz->bxsz', L, M)
Y  = np.einsum('bBst,bxsz->Bxtz', W, X)
HM = np.einsum('Bxtz,Bza->xta', Y, R)
return HM.reshape(Dl * d * Dr, order='C')

Nloc = Dl * d * Dr
Heff = LinearOperator(
shape=(Nloc, Nloc),
matvec=lambda v: heff_one_site_matvec(v, L, W, R, Dl, d, Dr),
dtype=np.complex128
)
```

The eigensolver should be initialized with the current local tensor:

```
v0 = M_current.reshape(Nloc, order='C')
```

D. Validation strategy

For small systems, one should explicitly build the dense projected operator

$$P^\dagger H P$$

for the one-site center and compare it against the dense matrix obtained by repeated application of the matrix-free routine above.

a. Important warning. Hermiticity of the environment-built local operator is necessary but not sufficient. With the conventions of these notes, a naive contraction using the stored left environment without the required index interchange can still produce a Hermitian operator, while failing to agree with the true projected operator $P^\dagger H P$. Therefore the contraction sequence in Eqs. (46)–(48) should be treated as the authoritative implementation for the present conventions.

X. TWO-SITE EFFECTIVE HAMILTONIAN

At bond $(i, i+1)$, define the two-site center tensor

$$\Theta^{[i,i+1]}_{\alpha_{i-1},\sigma_i,\sigma_{i+1},\alpha_{i+1}}, \tag{49}$$

so that

$$|\psi\rangle = \sum_{\alpha_{i-1},\sigma_i,\sigma_{i+1},\alpha_{i+1}} \Theta^{[i,i+1]}_{\alpha_{i-1},\sigma_i,\sigma_{i+1},\alpha_{i+1}} |\alpha_{i-1}\rangle_L |\sigma_i\rangle |\sigma_{i+1}\rangle |\alpha_{i+1}\rangle_R. \tag{50}$$

The two-site local variational problem is the direct analogue of the standard two-site DMRG superblock problem, but expressed entirely in terms of MPS/MPO environments and a matrix-free local action.

a. General structure. In principle, the effective Hamiltonian at bond $(i, i+1)$ may be written symbolically as a contraction of

- a left environment built from sites $1, \ldots, i-1$,
- the two local MPO tensors at sites i and $i+1$,
- a right environment built from sites $i+2, \ldots, L$.

In practice, however, the most reliable specification is again the validated contraction sequence itself.

b. Convention-sensitive point. For the conventions used in these notes, two implementation details are essential:

1. the stored left environment enters the local action with its two MPS-bond indices interchanged,
2. the local MPO tensors must be contracted with the physical-index routing that is consistent with the validated implementation below.

These points are easy to get wrong, and more naive contractions can produce Hermitian local operators that nevertheless fail to agree with the true projected operator

$$P^\dagger H P.$$

A. Validated matrix-free action

Let

$$L^{[i]}_{\beta_{i-1}}(\alpha_{i-1}, \alpha'_{i-1})$$

be the stored left environment built from sites $1, \ldots, i-1$, and let

$$R^{[i+1]}_{\beta_{i+1}}(\alpha_{i+1}, \alpha'_{i+1})$$

be the stored right environment built from sites $i+2, \ldots, L$.

The validated matrix-free two-site action is

$$X_{\beta_{i-1},\alpha_{i-1},\sigma_i,\sigma_{i+1},\alpha'_{i+1}} = \sum_{\alpha'_{i-1}} L^{[i]}_{\beta_{i-1}}(\alpha'_{i-1}, \alpha_{i-1}) \Theta_{\alpha'_{i-1},\sigma_i,\sigma_{i+1},\alpha'_{i+1}}, \tag{51}$$

$$Y_{\beta_i,\alpha_{i-1},\sigma'_i,\sigma_{i+1},\alpha'_{i+1}} = \sum_{\beta_{i-1},\sigma_i} W^{[i]}_{\beta_{i-1},\beta_i,\sigma_i,\sigma'_i} X_{\beta_{i-1},\alpha_{i-1},\sigma_i,\sigma_{i+1},\alpha'_{i+1}}, \tag{52}$$

$$Z_{\beta_{i+1},\alpha_{i-1},\sigma'_i,\sigma'_{i+1},\alpha'_{i+1}} = \sum_{\beta_i,\sigma_{i+1}} W^{[i+1]}_{\beta_i,\beta_{i+1},\sigma_{i+1},\sigma'_{i+1}} Y_{\beta_i,\alpha_{i-1},\sigma'_i,\sigma_{i+1},\alpha'_{i+1}}, \tag{53}$$

$$(\mathcal{H}_{\text{eff}}\Theta)_{\alpha_{i-1},\sigma'_i,\sigma'_{i+1},\alpha_{i+1}} = \sum_{\beta_{i+1},\alpha'_{i+1}} Z_{\beta_{i+1},\alpha_{i-1},\sigma'_i,\sigma'_{i+1},\alpha'_{i+1}} R^{[i+1]}_{\beta_{i+1}}(\alpha'_{i+1}, \alpha_{i+1}). \tag{54}$$

As in the one-site case, the first line uses

$$L^{[i]}_{\beta_{i-1}}(\alpha'_{i-1}, \alpha_{i-1})$$

rather than

$$L^{[i]}_{\beta_{i-1}}(\alpha_{i-1}, \alpha'_{i-1}).$$

B. Validated NumPy implementation

Let the two-site tensor be stored as

$$\Theta[\alpha_{i-1}, \sigma_i, \sigma_{i+1}, \alpha_{i+1}],$$

with shape

$$(D_{i-1}, d_i, d_{i+1}, D_{i+1}).$$

With arrays

- `L[b,x,y]` for the stored left environment,
- `Theta[y,u,v,z]` for the input two-site tensor,
- `W1[b,B,u,s]` for the MPO tensor on site i,
- `W2[B,C,v,t]` for the MPO tensor on site $i+1$,
- `R[C,z,a]` for the stored right environment,

the validated matrix-free contraction is

```
X  = np.einsum('byx,yuvz->bxuvz', L, Theta)
Y  = np.einsum('bBus,bxuvz->Bxsvz', W1, X)
Z  = np.einsum('BCvt,Bxsvz->Cxstz', W2, Y)
HT = np.einsum('Cxstz,Cza->xsta', Z, R)
```

The output tensor `HT[x,s,t,a]` has the same shape as `Theta`.

a. Interpretation of physical indices. The MPO tensors are stored as

$$W^{[i]}_{\beta_{i-1},\beta_i,\sigma,\sigma'},$$

with the first physical index the ket/output index and the second the bra/input index. The routing in the two-site contraction above is therefore convention-dependent and should be used exactly as written.

C. Vectorization and local eigensolver interface

If

$$\Theta^{[i,i+1]} \in \mathbb{C}^{D_{i-1}\times d_i \times d_{i+1}\times D_{i+1}},$$

then the local vector-space dimension is

$$N_{\text{loc}} = D_{i-1} d_i d_{i+1} D_{i+1}.$$

The eigensolver vector

$$v \in \mathbb{C}^{N_{\text{loc}}}$$

is identified with Θ by

$$\Theta = \text{reshape}(v,\ (D_{i-1}, d_i, d_{i+1}, D_{i+1})),$$

again using the fixed C-order convention of Section II.

A practical wrapper is

```
def heff_two_site_matvec(v, L, W1, W2, R, Dl, d1, d2, Dr):
Theta = v.reshape((Dl, d1, d2, Dr), order='C')
X  = np.einsum('byx,yuvz->bxuvz', L, Theta)
Y  = np.einsum('bBus,bxuvz->Bxsvz', W1, X)
Z  = np.einsum('BCvt,Bxsvz->Cxstz', W2, Y)
HT = np.einsum('Cxstz,Cza->xsta', Z, R)
return HT.reshape(Dl * d1 * d2 * Dr, order='C')
```

This routine is then passed to `LinearOperator` and solved by Lanczos or Davidson exactly as in the one-site case.

D. Validation strategy

For small systems, the dense matrix generated by repeated application of the matrix-free two-site routine above should be compared entrywise against the exact projected operator

$$P^\dagger H P.$$

This is the decisive correctness test for the environment-based implementation.

a. Important warning. Hermiticity alone is not a sufficient diagnostic. In the present conventions, one can obtain a two-site environment-built operator that is Hermitian and numerically stable, yet still incorrect because of an index-order mismatch in the left environment or in the MPO physical-index routing. Therefore the contraction sequence in Eqs. (51)–(54) should be treated as the authoritative implementation for the present conventions.

XI. MATRIX-FREE EIGENSOLVER INTERFACE FOR LOCAL EFFECTIVE HAMILTONIANS

In practical DMRG implementations, the local effective Hamiltonian should *never* be formed as a dense matrix except for unit tests on very small systems. Instead, one implements only the map

$$v \mapsto \mathcal{H}_{\text{eff}} v,$$

and passes this map to an iterative eigensolver such as Lanczos or Davidson.

a. Important convention warning. For the tensor, MPO, and environment conventions adopted in these notes, the stored left environment tensor enters the local effective-Hamiltonian action with its two MPS-bond indices interchanged. In array form, if the stored left environment is `L[b,x,y]`, then the validated local matrix-free contractions use it as `L[b,y,x]`. This point is subtle but essential. A local operator built from environments can remain Hermitian even when this ordering is wrong, while still failing to agree with the true projected operator

$$P^\dagger H P.$$

Accordingly, the one-site and two-site matrix-free contractions given below should be treated as the authoritative implementations for the conventions of these notes.

A. One-site local solve

For a one-site center tensor

$$M^{[i]} \in \mathbb{C}^{D_{i-1} \times d_i \times D_i},$$

define the local vector-space dimension

$$N_{\text{loc}} = D_{i-1} d_i D_i.$$

The eigensolver vector

$$v \in \mathbb{C}^{N_{\text{loc}}}$$

is identified with the tensor $M^{[i]}$ by

$$M^{[i]} = \text{reshape}(v,\, (D_{i-1}, d_i, D_i)), \tag{55}$$

using the fixed C-order convention of Section II. After applying the local effective Hamiltonian, the result is reshaped back to a vector with the same convention.

a. Validated one-site contraction. The one-site matrix-free action is the validated contraction defined in Section ??. In array form, with stored environment tensor `L[b,x,y]`, local tensor `M[y,s,z]`, MPO tensor `W[b,B,s,t]`, and right environment `R[B,z,a]`, the authoritative contraction is

```
X  = np.einsum('byx,ysz->bxsz', L, M)
Y  = np.einsum('bBst,bxsz->Bxtz', W, X)
HM = np.einsum('Bxtz,Bza->xta', Y, R)
```

b. Matrix-free wrapper. A practical NumPy/`scipy` interface is:

```
from scipy.sparse.linalg import LinearOperator, eigsh

def heff_one_site_matvec(v, L, W, R, Dl, d, Dr):
M = v.reshape((Dl, d, Dr), order='C')
```

```
X  = np.einsum('byx,ysz->bxsz', L, M)
Y  = np.einsum('bBst,bxsz->Bxtz', W, X)
HM = np.einsum('Bxtz,Bza->xta', Y, R)
return HM.reshape(Dl * d * Dr, order='C')

Nloc = Dl * d * Dr
Heff = LinearOperator(
shape=(Nloc, Nloc),
matvec=lambda v: heff_one_site_matvec(v, L, W, R, Dl, d, Dr),
dtype=np.complex128
)

E, vec = eigsh(Heff, k=1, which='SA', v0=v0)
M = vec.reshape((Dl, d, Dr), order='C')
```

Here `v0` should be the current local tensor flattened with the same reshape convention:

```
v0 = M_current.reshape(Nloc, order='C')
```

c. Hermiticity and correctness checks. For a Hermitian MPO and correctly built environments, the local operator should satisfy

$$\langle x, \mathcal{H}_{\text{eff}} y\rangle = \langle \mathcal{H}_{\text{eff}} x, y\rangle$$

up to numerical precision. However, Hermiticity alone is not a sufficient correctness test. For small systems, one should also compare the dense matrix obtained from the matrix-free routine against the exact projected operator

$$P^\dagger H P.$$

B. Two-site local solve

For a two-site center tensor

$$\Theta^{[i,i+1]} \in \mathbb{C}^{D_{i-1}\times d_i \times d_{i+1}\times D_{i+1}},$$

the local vector-space dimension is

$$N_{\text{loc}} = D_{i-1} d_i d_{i+1} D_{i+1}.$$

Again, the eigensolver vector v is identified with the tensor Θ by

$$\Theta = \text{reshape}(v,\ (D_{i-1}, d_i, d_{i+1}, D_{i+1})). \tag{56}$$

After applying the local effective Hamiltonian, the output tensor is flattened back to a vector using the same C-order convention.

a. Validated two-site contraction. The two-site matrix-free action is the validated contraction defined in Section **??**. In array form, with stored left environment `L[b,x,y]`, two-site tensor `Theta[y,u,v,z]`, MPO tensors `W1[b,B,u,s]` and `W2[B,C,v,t]`, and stored right environment `R[C,z,a]`, the authoritative contraction is

```
X  = np.einsum('byx,yuvz->bxuvz', L, Theta)
Y  = np.einsum('bBus,bxuvz->Bxsvz', W1, X)
Z  = np.einsum('BCvt,Bxsvz->Cxstz', W2, Y)
HT = np.einsum('Cxstz,Cza->xsta', Z, R)
```

b. Matrix-free wrapper. A practical wrapper is

```
def heff_two_site_matvec(v, L, W1, W2, R, Dl, d1, d2, Dr):
Theta = v.reshape((Dl, d1, d2, Dr), order='C')
X  = np.einsum('byx,yuvz->bxuvz', L, Theta)
Y  = np.einsum('bBus,bxuvz->Bxsvz', W1, X)
Z  = np.einsum('BCvt,Bxsvz->Cxstz', W2, Y)
HT = np.einsum('Cxstz,Cza->xsta', Z, R)
return HT.reshape(Dl * d1 * d2 * Dr, order='C')
```

and this is passed to `LinearOperator` in the same way as for the one-site case.

c. Practical remark. The two-site contraction above was explicitly validated against the dense projected operator

$$P^\dagger H P$$

for small systems. This validation is essential, because more naive einsum routings can produce Hermitian local operators that are nevertheless incorrect.

C. Lanczos versus Davidson

For first implementations, Lanczos is usually sufficient and easiest to access through standard sparse-eigensolver interfaces. Davidson can be advantageous when the local dimension becomes larger or when a useful diagonal preconditioner is available.

- **Lanczos:** simplest robust choice for initial code.
- **Davidson:** often faster for large local problems, but requires more implementation effort.

In both cases, using the current local tensor as the initial guess is essential for sweep efficiency.

D. Dense reference for unit tests

For very small dimensions only, one should explicitly build the dense local projected operator

$$H_{\text{eff}} \in \mathbb{C}^{N_{\text{loc}} \times N_{\text{loc}}},$$

and compare:

- dense matrix–vector multiplication versus the matrix-free routine,
- lowest eigenvalue from dense diagonalization versus Lanczos/Davidson,
- Hermiticity of the dense reference matrix,
- agreement of the environment-built operator with the exact projected operator $P^\dagger H P$.

This unit test should be performed for both the one-site and two-site local solvers before running full sweeps.

XII. TWO-SITE DMRG

Two-site DMRG is the most robust finite-system ground-state algorithm and is recommended for a first implementation. It avoids many local-minimum problems of one-site DMRG and dynamically adjusts the bond dimension before truncation.

A. Two-site mixed-canonical form

At bond $(i, i+1)$, define left-orthonormal Schmidt states on sites $1, \ldots, i-1$ and right-orthonormal Schmidt states on sites $i+2, \ldots, L$:

$$|\alpha_{i-1}\rangle_L, \qquad |\alpha_{i+1}\rangle_R.$$

The variational two-site center tensor is

$$\Theta^{[i,i+1]}_{\alpha_{i-1},\sigma_i,\sigma_{i+1},\alpha_{i+1}}, \tag{57}$$

and the state is written as

$$|\psi\rangle = \sum_{\alpha_{i-1},\sigma_i,\sigma_{i+1},\alpha_{i+1}} \Theta^{[i,i+1]}_{\alpha_{i-1},\sigma_i,\sigma_{i+1},\alpha_{i+1}} |\alpha_{i-1}\rangle_L |\sigma_i\rangle |\sigma_{i+1}\rangle |\alpha_{i+1}\rangle_R. \tag{58}$$

This form is the direct analogue of the usual two-site DMRG superblock state and should be treated as the primary variational object. It is not necessary to define Θ through pre-existing tensors $A^{[i]}$ and $B^{[i+1]}$.

B. Solving the local problem

Given a matrix-free routine for $\Theta \mapsto \mathcal{H}_{\text{eff}}\Theta$, solve for the lowest eigenpair using Lanczos or Davidson:

$$\mathcal{H}_{\text{eff}}\Theta = E\Theta. \tag{59}$$

The local vector space dimension is

$$N_{\text{loc}} = D_{i-1} d_i d_{i+1} D_{i+1}. \tag{60}$$

For spin-1/2 chains with uniform physical dimension $d_i = d_{i+1} = 2$, this is

$$N_{\text{loc}} = 4 D_{i-1} D_{i+1}.$$

The eigensolver should accept an initial guess. A good choice is the current two-site center tensor before optimization, flattened with the same C-order convention used everywhere else. This substantially accelerates convergence once sweeps are underway.

C. SVD split and truncation

After optimization, reshape the two-site tensor as a matrix:

$$\Theta_{(\alpha_{i-1}\sigma_i),(\sigma_{i+1}\alpha_{i+1})} \in \mathbb{C}^{(D_{i-1}d_i)\times(d_{i+1}D_{i+1})}. \tag{61}$$

Compute the reduced SVD:

$$\Theta = USV^\dagger. \tag{62}$$

Let

$$D_{\text{full}} = \text{rank}(\Theta), \qquad D_{\text{new}} = \min(D_{\max}, D_{\text{full}})$$

or more generally choose D_{new} from a singular-value threshold or discarded-weight tolerance. Keep only the largest D_{new} singular values.

The discarded weight is

$$\eta^{[i]} = \sum_{j > D_{\text{new}}} s_j^2. \tag{63}$$

For a left-to-right sweep, define

$$A^{[i]} \leftarrow \text{reshape}(U,\, (D_{i-1}, d_i, D_{\text{new}}))\,, \tag{64}$$

$$M^{[i+1]} \leftarrow \text{reshape}\big(SV^\dagger,\, (D_{\text{new}}, d_{i+1}, D_{i+1})\big)\,. \tag{65}$$

Then $A^{[i]}$ is left-canonical, and the orthogonality center has moved to site $i+1$.

For a right-to-left sweep, instead use

$$M^{[i]} \leftarrow \text{reshape}(US,\, (D_{i-1}, d_i, D_{\text{new}}))\,, \tag{66}$$

$$B^{[i+1]} \leftarrow \text{reshape}\big(V^\dagger,\, (D_{\text{new}}, d_{i+1}, D_{i+1})\big)\,, \tag{67}$$

so that $B^{[i+1]}$ is right-canonical and the orthogonality center moves to site i.

a. Important reshape note. The reshapes in Eqs. (64)–(67) are only correct if Eq. (61) uses exactly the same flattening convention as the code.

D. Environment updates

After splitting the optimized two-site tensor, the environments must be updated so that the next local problem is built in the correct mixed-canonical gauge.

a. Left-to-right sweep. After obtaining the new left-canonical tensor $A^{[i]}$, update the left environment from site i to site $i+1$:

$$L^{[i+1]}_{\beta_i}(\alpha_i, \alpha'_i) = \sum_{\substack{\alpha_{i-1},\alpha'_{i-1}\\ \sigma_i,\sigma'_i\\ \beta_{i-1}}} L^{[i]}_{\beta_{i-1}}(\alpha_{i-1}, \alpha'_{i-1}) A^{[i]}_{\alpha_{i-1},\sigma_i,\alpha_i} W^{[i]}_{\beta_{i-1},\beta_i,\sigma_i,\sigma'_i} A^{[i]*}_{\alpha'_{i-1},\sigma'_i,\alpha'_i}. \tag{68}$$

b. Right-to-left sweep. After obtaining the new right-canonical tensor $B^{[i+1]}$, update the right environment from site $i+1$ to site i:

$$R^{[i]}_{\beta_i}(\alpha_i, \alpha'_i) = \sum_{\substack{\alpha_{i+1},\alpha'_{i+1}\\ \sigma_{i+1},\sigma'_{i+1}\\ \beta_{i+1}}} B^{[i+1]}_{\alpha_i,\sigma_{i+1},\alpha_{i+1}} W^{[i+1]}_{\beta_i,\beta_{i+1},\sigma_{i+1},\sigma'_{i+1}} R^{[i+1]}_{\beta_{i+1}}(\alpha_{i+1}, \alpha'_{i+1}) B^{[i+1]*}_{\alpha'_i,\sigma'_{i+1},\alpha'_{i+1}}. \tag{69}$$

c. Convention reminder. The formulas above define the *stored* environments. As emphasized in Section VIII, for the conventions of these notes the stored left environment enters the local effective-Hamiltonian action with its two MPS-bond indices interchanged relative to its storage order. Thus the environment-recursion formulas and the local-action formulas must not be conflated.

E. Full two-site sweep algorithm

Algorithm 3 TwoSiteDMRG

Require: MPO $\{W^{[i]}\}$, initial MPS $\{M^{[i]}\}$, max bond dimension $D_{\max}$
Ensure: approximate ground-state MPS and energy

```
bring initial MPS into mixed-canonical form
build all right environments R^[i]
initialize left boundary environment L^[1]
repeat                                                    ▷ Left-to-right sweep
    for i = 1 to L − 1 do
        form the current two-site center tensor Θ^[i,i+1]
        solve the lowest-eigenvalue problem H_eff Θ = E Θ
        reshape Θ as in Eq. (61)
        compute reduced SVD Θ = U S V†
        truncate to D_new ≤ D_max
        set A^[i] ← reshape(U, (D_{i−1}, d_i, D_new))
        set M^[i+1] ← reshape(S V†, (D_new, d_{i+1}, D_{i+1}))
        record discarded weight η^[i]
        update left environment L^[i+1]
    end for                                               ▷ Right-to-left sweep
    for i = L − 1 down to 1 do
        form the current two-site center tensor Θ^[i,i+1]
        solve the lowest-eigenvalue problem
        reshape Θ as in Eq. (61)
        compute reduced SVD Θ = U S V†
        truncate to D_new ≤ D_max
        set M^[i] ← reshape(U S, (D_{i−1}, d_i, D_new))
        set B^[i+1] ← reshape(V†, (D_new, d_{i+1}, D_{i+1}))
        record discarded weight η^[i]
        update right environment R^[i]
    end for
until energy, variance, and discarded-weight criteria are converged
```

F. Practical comments for coding

- **This should be the first DMRG implementation.** It is much easier to debug than one-site DMRG.
- **Keep the orthogonality center explicit.** During a left-to-right sweep, after the split the center lives on site $i+1$ as $M^{[i+1]}$; during a right-to-left sweep, it lives on site i as $M^{[i]}$.
- **Do not compare tensors directly across sweeps without gauge fixing.**
- **Always unit test the local effective-Hamiltonian matvec** against a dense reference for small dimensions.
- **Monitor the maximum discarded weight**

$$\eta_{\max} = \max_i \eta^{[i]}$$

  during each sweep.
- **Benchmark first against exact diagonalization** for small Heisenberg chains before attempting larger systems.

XIII. ONE-SITE DMRG

One-site DMRG is best treated as an advanced refinement of the two-site algorithm. In practice, a robust implementation typically uses several two-site sweeps first and only then switches to one-site sweeps, optionally with subspace expansion or density-matrix perturbation.

A. One-site mixed-canonical form

In one-site DMRG, the MPS is kept in mixed-canonical form with center at site i:

$$|\psi\rangle = \sum_{\alpha_{i-1},\sigma_i,\alpha_i} M^{[i]}_{\alpha_{i-1},\sigma_i,\alpha_i} |\alpha_{i-1}\rangle_L |\sigma_i\rangle |\alpha_i\rangle_R. \tag{70}$$

The local optimization is then performed over the center tensor $M^{[i]}$ while the left and right Schmidt bases are kept fixed.

B. One-site sweep algorithm

Algorithm 4 OneSiteDMRG

Require: MPO $\{W^{[i]}\}$, initial MPS $\{M^{[i]}\}$, max bond dimension $D_{\max}$
Ensure: approximate ground-state MPS and energy

```
bring the initial MPS into mixed-canonical form
build all right environments
initialize the left boundary environment
repeat                                                          ▷ Left-to-right sweep
    for i = 1 to L − 1 do
        solve the one-site effective-Hamiltonian problem using the validated matrix-free contraction of Section XI
        reshape the optimized center tensor as a left-grouped matrix
        compute an SVD: M^[i] = USV^†
        retain at most D_max singular values
        reshape U → A^[i] as a left-canonical tensor
        absorb SV^† into M^[i+1]
        update L^[i+1]
    end for                                                     ▷ Right-to-left sweep
    for i = L down to 2 do
        solve the one-site effective-Hamiltonian problem using the validated matrix-free contraction of Section XI
        reshape the optimized center tensor as a right-grouped matrix
        compute an SVD: M^[i] = USV^†
        retain at most D_max singular values
        reshape V^† using the inverse of Eq. (2) to obtain a right-canonical tensor B^[i]
        absorb US into M^[i−1]
        update R^[i−1]
    end for
until energy and variance are converged
```

a. Remark. In standard one-site DMRG, the SVD step mainly regauges the state and shifts the orthogonality center; unlike two-site DMRG, it does not naturally enlarge the accessible bond space. This is one reason why one-site DMRG is more prone to local minima unless combined with subspace expansion or density-matrix perturbation.

b. Convention reminder. For the conventions of these notes, the one-site local matrix-free action used inside one-site DMRG must be the validated contraction summarized in Section XI, rather than a naive contraction built directly from the stored left environment without the required bond-index interchange.

c. Practical warning. One-site DMRG is prone to local minima and should not be used as a first standalone implementation. A robust implementation should either:

- start with several two-site sweeps, or
- use a subspace-expansion / density-matrix perturbation method.

XIV. AKLT BENCHMARK STATE

An exact MPS representation of the AKLT state is, up to gauge, given by

$$A^{+} = \begin{pmatrix} 0 & \sqrt{2/3} \\ 0 & 0 \end{pmatrix}, \tag{71}$$

$$A^{0} = \begin{pmatrix} -1/\sqrt{3} & 0 \\ 0 & 1/\sqrt{3} \end{pmatrix}, \tag{72}$$

$$A^{-} = \begin{pmatrix} 0 & 0 \\ -\sqrt{2/3} & 0 \end{pmatrix}. \tag{73}$$

For this gauge, the tensors satisfy

$$\sum_{\sigma\in\{+,0,-\}} A^{\sigma} A^{\sigma\dagger} = \mathbb{I}_2, \tag{74}$$

so they are *right-canonical* in the convention of Eq. (6). This state is an excellent benchmark:

- the AKLT ground state admits an exact MPS representation with bond dimension $D = 2$,
- for open boundary conditions, the AKLT Hamiltonian has a fourfold degenerate ground-state manifold associated with spin-1/2 edge modes, and every ground state in that manifold has exact energy

$$E_0 = -\frac{2}{3}(L-1).$$

XV. CONVERGENCE DIAGNOSTICS

Reliable convergence diagnostics for DMRG should include:

- energy change

$$|\Delta E| < \epsilon_E,$$

- energy variance

$$\sigma_H^2 = \langle\psi|H^2|\psi\rangle - \langle\psi|H|\psi\rangle^2,$$

- maximum discarded weight during a sweep,
- stability of entanglement spectrum or local observables.

a. Practical note. For small systems, the variance can be benchmarked against dense linear algebra. In production calculations, it can be obtained either from an MPO representation of H^2 or by first computing $H|\psi\rangle$ as an MPS and then evaluating its norm. One should keep in mind that an MPO for H^2 can have a significantly larger bond dimension than the MPO for H.

b. Important caution. Direct tensor-by-tensor comparison is not gauge-invariant and should *not* be used as a primary convergence criterion unless the gauge is fixed identically before comparison.

XVI. IMPLEMENTATION CHECKLIST

Before production calculations, the implementation should be validated in a strictly controlled order. The purpose of this checklist is not only to test individual routines, but also to ensure that the tensor-index conventions, reshape conventions, and environment contractions are mutually consistent.

1. **Canonicalization test.** Starting from random tensors or a random MPS, perform left- and right-canonicalization sweeps and verify the defining relations

$$\sum_{\alpha_{i-1},\sigma_i} A^{[i]}_{\alpha_{i-1},\sigma_i,\alpha_i} A^{[i]*}_{\alpha_{i-1},\sigma_i,\alpha'_i} = \delta_{\alpha_i,\alpha'_i}, \tag{75}$$

$$\sum_{\sigma_i,\alpha_i} B^{[i]}_{\alpha_{i-1},\sigma_i,\alpha_i} B^{[i]*}_{\alpha'_{i-1},\sigma_i,\alpha_i} = \delta_{\alpha_{i-1},\alpha'_{i-1}}. \tag{76}$$

   These tests verify both the QR/SVD logic and the fixed reshape conventions.

2. **MPO construction test.** For small chains, such as $L = 2, 3, 4$, contract the MPO to a dense matrix and compare it entrywise against a Hamiltonian built directly from Kronecker products of local operators. This test should be performed for both the Heisenberg MPO and the AKLT MPO.

3. **Environment-recursion test.** Verify that the left and right environment updates reproduce brute-force contractions on very small systems. This confirms that the recursive environment-building routines are internally consistent.

4. **Hermiticity test of local matrix-free operators.** For random small tensors, verify numerically that

$$\langle x, \mathcal{H}_{\text{eff}} y\rangle = \langle \mathcal{H}_{\text{eff}} x, y\rangle$$

   up to numerical precision, for both one-site and two-site matrix-free local actions.

5. **One-site projected-operator test.** For a small mixed-canonical MPS, explicitly construct the dense projected one-site operator

$$P^\dagger H P$$

   and compare it against the dense matrix obtained by repeated application of the matrix-free one-site environment-based local action.

   For the conventions of these notes, the validated one-site contraction is

```
X  = np.einsum('byx,ysz->bxsz', L, M)
Y  = np.einsum('bBst,bxsz->Bxtz', W, X)
HM = np.einsum('Bxtz,Bza->xta', Y, R)
```

6. **Two-site projected-operator test.** For a small mixed-canonical MPS, explicitly construct the dense projected two-site operator

$$P^\dagger H P$$

   and compare it against the dense matrix obtained by repeated application of the matrix-free two-site environment-based local action.

   For the conventions of these notes, the validated two-site contraction is

```
X  = np.einsum('byx,yuvz->bxuvz', L, Theta)
Y  = np.einsum('bBus,bxuvz->Bxsvz', W1, X)
Z  = np.einsum('BCvt,Bxsvz->Cxstz', W2, Y)
HT = np.einsum('Cxstz,Cza->xsta', Z, R)
```

   This comparison against the dense projected operator is the decisive local correctness test.

7. **Two-site DMRG benchmark for Heisenberg chains.** Run finite-system two-site DMRG for small spin-1/2 Heisenberg chains and compare the converged MPS energy against exact diagonalization. Agreement to numerical precision on small systems is the main global benchmark for the implementation.

8. **AKLT benchmark.** Validate the AKLT implementation in two ways:

- compare the DMRG ground-state energy against the exact open-chain result

$$E_0 = -\frac{2}{3}(L-1),$$

- evaluate the energy of the exact AKLT MPS and verify that it reproduces the same result.

9. **Convergence diagnostics in production runs.** After the implementation is validated locally and globally, monitor in larger calculations:

   - energy change between sweeps,
   - maximum discarded weight,
   - eventual energy variance,
   - stability of local observables and entanglement spectra.

a. Most important practical warning. For the conventions used in these notes, Hermiticity of the environment-built local operator is *not* sufficient to guarantee correctness. In particular, an incorrect treatment of the stored left environment or an incorrect MPO physical-index routing can still yield a Hermitian local operator, yet one that differs from the true projected operator $P^\dagger HP$. Therefore no full DMRG sweeps should be trusted until the one-site and two-site local operators have been compared explicitly against dense projected references on small systems.

b. Recommended debugging order. If a full DMRG calculation produces stable but incorrect energies, the safest debugging sequence is:

1. verify canonicalization,
2. verify MPO-to-dense agreement,
3. verify left and right environment updates separately,
4. verify the one-site local projected operator,
5. verify the two-site local projected operator,
6. only then test full sweeps.

In practice, the projected-operator tests are usually more informative than Hermiticity tests alone.

c. Authoritative implementation templates. For the conventions of these notes, the one-site and two-site matrix-free contraction templates written above should be treated as authoritative reference implementations. Future optimizations, refactorings, or changes of contraction order should always be benchmarked back against these forms on small systems before being used in production calculations.

XVII. PRACTICAL CODING RECOMMENDATION

For a first reliable implementation, the following strategy is strongly recommended.

- **Implement two-site DMRG first.** Among standard finite-system algorithms, two-site DMRG is the most robust place to begin. It is substantially easier to debug than one-site DMRG, naturally allows bond dimensions to adapt before truncation, and provides a direct setting in which the local variational space is large enough to avoid many of the local-minimum problems of one-site updates.
- **Fix tensor conventions once and use them everywhere.** In particular, fix:
  - the MPS tensor convention

$$A^{[i]}_{\alpha_{i-1},\sigma_i,\alpha_i},$$

  - the MPO tensor convention

$$W^{[i]}_{\beta_{i-1},\beta_i,\sigma_i,\sigma_i'},$$

    with the first physical index the ket/output index and the second the bra/input index,

  - the reshape rules used in QR, SVD, vectorization, and inverse reshaping,
  - the memory order, e.g. NumPy C-order.

  Small inconsistencies in these choices are among the most common sources of silent errors.

- **Use reduced QR for regauging and SVD for truncation.** Reduced QR without pivoting is convenient for moving orthogonality centers and constructing canonical forms. SVD should be used when a truncation decision is required, because it gives direct access to Schmidt singular values and discarded weights.

- **Keep the orthogonality-center structure explicit.** In two-site DMRG, the variational object is the two-site tensor

$$\Theta^{[i,i+1]}_{\alpha_{i-1},\sigma_i,\sigma_{i+1},\alpha_{i+1}},$$

  and the tensors to the left and right should be kept in the corresponding mixed-canonical form. During a left-to-right sweep, after splitting the optimized two-site tensor, the center moves to the right; during a right-to-left sweep, it moves to the left. This structure should be reflected explicitly in the code.

- **Do not build dense local effective Hamiltonians in production sweeps.** Except for very small validation tests, the local effective Hamiltonian should be accessed only through a matrix-free map

$$v \mapsto \mathcal{H}_{\text{eff}} v.$$

  This is the only scalable strategy and is the natural interface to Lanczos or Davidson.

- **Treat the validated contraction templates as authoritative.** For the conventions of these notes, the one-site and two-site local environment-based contractions are subtle. In particular, the stored left environment enters the local action with its MPS-bond indices interchanged, and the MPO physical-index routing in the two-site action must be implemented exactly as in the validated contraction sequence. These convention-dependent details should not be re-derived from memory during coding.

- **Benchmark against dense projected operators before full sweeps.** Before running finite-system DMRG sweeps, one should verify on small systems that the environment-built one-site and two-site local operators agree with the exact projected operators

$$P^\dagger H P.$$

  This is a more stringent and more informative test than Hermiticity alone.

- **Benchmark against exact diagonalization on small systems.** After the local-operator tests pass, run two-site DMRG for small Heisenberg chains and compare the converged energy against exact diagonalization. This is the standard global correctness test.

- **Use the AKLT chain as an exact MPS benchmark.** The AKLT model is particularly valuable because:

  - the Hamiltonian has an exact MPO representation of modest bond dimension,
  - the ground state admits an exact MPS representation with bond dimension $D = 2$,
  - the open-chain ground-state energy is known exactly,

$$E_0 = -\frac{2}{3}(L-1).$$

  Therefore the AKLT chain tests simultaneously the MPO, the local solver, the sweep logic, and the handling of exact low-bond-dimension MPS ground states.

- **Use the current local tensor as the eigensolver initial guess.** In finite-system sweeps, the previously optimized local tensor provides an excellent initial vector for Lanczos or Davidson. This is a major practical improvement in convergence speed.

- **Monitor discarded weights and, when available, variances.** The energy alone is not always a sufficient convergence indicator. In production calculations, one should monitor at least:

  - the energy change between sweeps,
  - the maximum discarded weight during each sweep,
  - the eventual energy variance,
  - stability of local observables and entanglement spectra.

a. Recommended order of development. A robust development sequence is:

1. local operators,
2. MPO construction,
3. canonicalization routines,
4. environment updates,
5. one-site projected-operator validation,
6. two-site projected-operator validation,
7. two-site DMRG benchmark for Heisenberg chains,
8. AKLT benchmark,
9. only afterward: one-site DMRG, variance evaluation, and further optimizations.

b. Most important practical warning. For the conventions used in these notes, a local operator built from environments can be Hermitian and numerically stable while still being incorrect. This can happen if the stored left environment is inserted into the local action with the wrong bond-index order, or if the MPO physical-index routing in the local contraction is inconsistent with the chosen MPO storage convention. Therefore full DMRG sweeps should *not* be trusted until the matrix-free local operator has been compared explicitly against the dense projected operator $P^\dagger HP$ on small systems.

c. Final recommendation. In practical coding work, the safest philosophy is:

> first validate the local operator, then validate small-system energies, and only then scale up.

For environment-based DMRG implementations, this order is not merely convenient; it is essential.